AI Document Generator + Humanizer Agent

Complete Working Plan, Workflow, Architecture, Features, Database, Security, Document Generation, Humanization, Testing & Deployment

Purpose: A web-based AI system that creates professional documents from user-provided information, allows editing and rewriting, formats the final document, and exports it as PDF/DOCX.

Important design principle: The system should generate original, useful writing based on the user's real information. 'Humanize' means improving naturalness, clarity, flow and personalization—not guaranteeing that a third-party AI detector will be bypassed.

1. Project ka basic concept

Is project ka main idea ek AI-powered document platform banana hai jahan user sirf apni requirements/details de aur system us information ko process karke complete professional document bana de.

Example: user kahe: 'Mujhe digital marketing agency ke liye business proposal chahiye'. Agent questions/inputs collect karega, proposal generate karega, phir user preview karega aur PDF/DOCX export kar sakega.

2. User ke liye complete workflow

- User website/app open karega.
- Document type select karega.
- Required information form ya chat ke through provide karega.
- AI input ko analyze karke missing information identify karega.
- AI document ka outline/structure banayega.
- AI complete draft generate karega.
- Quality pass: grammar, consistency, duplicate/repetitive content aur formatting check.
- Humanize/Rewrite option se selected text ya complete document ko natural, clear aur audience-appropriate banaya ja sakta hai.
- User editor mein content manually edit karega.
- User template, font, spacing aur branding customize karega.
- System final PDF/DOCX generate karega.
- Document download/save/share kiya ja sakta hai.

3. Document types

Category	Examples
Career	CV/Resume, Cover Letter, Job Application, Recommendation Letter
Business	Business Proposal, Company Profile, Quotation, Sales Proposal, SOP
Academic	Report, Research Outline, Project Report, Presentation Content
Marketing	Blog, Product Description, Ad Copy, Email Campaign, Landing Page Copy
Office	Official Letter, Meeting Summary, Memo, Notice, Email
General	Article, Personal Statement, Statement of Purpose, Custom Document

4. AI agent ka actual kaam kaise hoga

Step A — Input collection

User ko sirf ek blank text box na dein. Har document ke liye smart form banayein. Example CV ke liye name, contact, education, experience, skills, projects, certifications aur target job role.

Step B — Requirement analysis

AI pehle user input ko structured data mein convert karega. Is stage par model document type, audience, tone, length aur required sections identify karega.

Step C — Missing information

Agar important information missing ho to agent targeted questions pooch sakta hai. Example: 'Company ka naam kya hai?' ya 'Proposal ki estimated budget range kya hai?'

Step D — Outline

AI pehle outline create karega. User chahe to outline approve/edit kare. Isse long documents mein hallucination aur irrelevant content kam hota hai.

Step E — Draft generation

Approved outline ke according final text generate hoga. AI ko system prompt mein document rules, tone, audience, format aur factual constraints diye jayenge.

Step F — Review

Second AI pass spelling, grammar, consistency, repeated ideas, headings, dates, names aur unsupported claims check karega.

Step G — Humanize

Humanizer writing ko natural aur audience-appropriate banayega: unnecessary repetition reduce karega, sentence length vary karega, awkward phrases improve karega aur user's actual voice/preferences ko preserve karega.

Step H — Export

Final structured content ko DOCX/PDF renderer ko diya jayega. Export ke waqt headers, footers, page numbers, tables, margins, logo aur typography apply hogi.

5. Humanizer ka recommended design

Humanizer ko ek single 'magic prompt' ke bajaye controlled rewrite pipeline banana zyada reliable hai.

- Original text ko preserve as source input.
- Meaning preservation check.
- Unnatural/repetitive phrases identify.
- Sentence flow and transitions improve.
- Tone selection: Professional, Friendly, Academic, Simple, Persuasive.
- Vocabulary ko audience ke level ke mutabiq adjust.
- Facts, names, numbers, URLs aur dates ko bina user permission change na karna.
- Final comparison: original vs rewritten version.
- User ko Accept / Reject / Regenerate controls dena.

```
Humanize Pipeline
Original Text
↓
Meaning & Facts Lock
↓
Style Analysis
```

```
↓
Natural Rewrite
↓
Consistency Check
↓
Final Humanized Text
```

6. Recommended website pages

- Landing Page — product explanation, examples, templates, pricing and CTA.
- Dashboard — recent documents, templates, usage and create-new button.
- Create Document — document type + guided form/chat.
- AI Editor — rich text editor with Generate, Rewrite, Humanize, Expand, Shorten and Improve buttons.
- Templates — professionally designed templates.
- Preview — exact final document preview before export.
- Export — PDF/DOCX options and file settings.
- Document History — previous versions and restore.
- Account/Settings — profile, preferences, language and usage.
- Admin Panel — users, templates, AI usage, logs, subscriptions/limits and abuse controls.

7. Recommended tech stack

Layer	Recommended technology
Frontend	Next.js + React + TypeScript + Tailwind CSS
Editor	Tiptap or Lexical
Backend	Next.js API routes / server actions or separate Node.js API
Database	Supabase PostgreSQL
Authentication	Supabase Auth
AI	LLM API with structured JSON output
File Storage	Supabase Storage or equivalent object storage
PDF	Server-side HTML/CSS to PDF renderer
DOCX	docx generation library
Deployment	Vercel for frontend/backend where suitable
Monitoring	Error tracking + server logs + usage analytics

8. Database structure

Supabase/PostgreSQL mein basic schema kuch is tarah ho sakta hai:

```
users
- id
- email
- name
- created_at

documents
- id
- user_id
- title
- document_type
- status
- current_version_id
- created_at
- updated_at

document_versions
- id
- document_id
- version_number
- content_json
- plain_text
- created_at

templates
- id
- name
- category
- template_json
- preview_url

ai_usage
- id
- user_id
- document_id
- operation
- input_tokens
- output_tokens
- estimated_cost
- created_at

exports
- id
- document_id
- format
- storage_path
- created_at
```

9. AI prompt architecture

AI ko multiple roles/stages mein divide karna useful hai. Example:

```
SYSTEM: You are a professional document generation assistant.
Rules:
1. Use only user-provided facts unless explicitly asked to create examples.
2. Do not invent qualifications, employment, dates, awards, statistics or certifications.
3. Preserve names, numbers, URLs and important factual details.
```

```
4. Follow the selected document type and tone.
5. Return structured JSON for the application to render.

USER:
Document type: Business Proposal
Audience: Small business owner
Tone: Professional
Length: 4-6 pages
User information: ...
```

10. Structured AI output

Raw AI text ko directly UI mein use karne ke bajaye JSON structure lena better hai:

```
{
"title": "Business Proposal",
"sections": [
{
"heading": "Executive Summary",
"paragraphs": ["..."]
},
{
"heading": "Services",
"bullets": ["...", "..."]
}
],
"warnings": []
}
```

11. Safety aur factual accuracy

- AI ko fake degrees, fake companies, fake experience, fake certificates ya fabricated references create nahi karne chahiye.
- User-provided facts ko preserve karna aur unknown information ko clearly mark karna chahiye.
- Legal, medical aur financial documents mein disclaimer/review workflow rakhna chahiye.
- Sensitive user documents encrypted storage aur access controls ke saath handle karein.
- Public document links ko random unguessable IDs aur expiry options ke saath protect karein.
- User ko delete-document option aur account/data deletion workflow dein.

12. Cost control

AI document platform mein sabse important operational cost LLM usage hai. Isliye:

- Short input par smaller/cheaper model use karein.
- Long documents ke liye chunking aur section-based generation use karein.
- Repeated instructions ko server-side/system prompt mein optimize karein.
- Humanize sirf selected section par run karne ka option dein.
- Per-user daily/monthly usage limits rakhein.
- Usage meter dashboard mein show karein.

- Generated documents ko unnecessary baar regenerate na karein.

13. Complete backend request flow

```
POST /api/documents
↓
Authenticate user
↓
Validate input
↓
Check usage/quota
↓
Create document record
↓
AI analysis
↓
Generate outline
↓
Generate sections
↓
Quality/fact check
↓
Save version
↓
Return structured document
↓
Editor / Preview

POST /api/documents/:id/humanize
↓
Load selected text
↓
Preserve facts
↓
Rewrite
↓
Quality check
↓
Save new version
```

14. PDF/DOCX generation

Best approach: application ke andar ek canonical document JSON rakhein. PDF aur DOCX dono isi same source se generate hon. Isse PDF aur Word versions ke content mein mismatch kam hota hai.

- Template JSON/HTML structure define karein.
- User branding: logo, company name, colors, contact details.
- A4/Letter page size option.
- Margins and typography settings.
- Header/footer and page numbers.
- Tables and bullet lists.
- Page-break controls.

- Final render preview.
- Generate PDF/DOCX and store securely.

15. Admin panel

- Total users and active users.
- Documents generated per day/month.
- AI operations and token usage.
- Top document types.
- Template management.
- User usage limits.
- Failed generation logs.
- Abuse/rate-limit monitoring.
- System prompt/version management.
- Export/download logs.

16. Testing checklist

Area	Test
AI	Correct structure, no fabricated facts, stable tone
Humanizer	Meaning preserved, names/numbers unchanged
Editor	Save, undo/redo, autosave, version restore
PDF	Fonts, page breaks, tables, headers, footer
DOCX	Open correctly in Microsoft Word
Auth	Unauthorized users cannot access documents
Security	File URLs, API keys and database rules protected
Performance	Large document generation handled without timeout
Mobile	Editor/dashboard usable on mobile
Failure	AI/API timeout gives retry without losing draft

17. MVP — pehle version mein kya banana hai

Agar aap jaldi launch karna chahte hain to pehle version ko unnecessarily huge na banayein.

- User signup/login.
- 5–10 document types.
- Smart input form.
- AI document generation.
- Rich text editor.

- Humanize/rewrite selected text.
- 3–5 professional templates.
- PDF export.
- DOCX export.
- Document history.
- Basic usage limits.
- Admin panel.

18. Phase 2 features

- Chat-based document creation.
- AI-powered template recommendation.
- Multi-language documents.
- Brand kit.
- Team/workspace collaboration.
- Comments and approval workflow.
- Document sharing links.
- Advanced analytics.
- API access.
- Subscription/billing system.
- Custom company templates.

19. Recommended user experience

User ko 20 fields ka form dene ke bajaye wizard/chat hybrid approach rakhein:

```
Create Document
↓
Choose Type
↓
Tell AI what you need
↓
AI asks only necessary questions
↓
Generate Outline
↓
Generate Draft
↓
Edit / Humanize / Improve
↓
Choose Template
↓
Preview
↓
Download PDF / DOCX
```

20. Deployment plan

- GitHub repository create karein.
- Next.js project ko Vercel se connect karein.
- Supabase project create karke database/auth/storage configure karein.
- AI provider API key ko Vercel environment variables mein rakhein.
- Production domain connect karein.
- Database Row Level Security (RLS) enable karein.
- Server-side AI calls use karein; API key browser mein expose na karein.
- Error logging and analytics setup karein.
- Production test documents generate karke PDF/DOCX verify karein.

21. Suggested project folder structure

```
app/
dashboard/
create/
editor/[id]/
templates/
preview/[id]/
api/
documents/
humanize/
export/
components/
editor/
templates/
forms/
preview/
lib/
ai/
db/
documents/
pdf/
docx/
validation/
supabase/
migrations/
types/
document.ts
template.ts
```

22. Client ko project explain karne ka short version

One-line explanation: “Ye AI-powered document platform hai jo user ki requirements se professional documents generate karta hai, unhein natural and polished writing mein improve karta hai, aur final document ko branded PDF/DOCX format mein export karta hai.”

Business use case: companies, freelancers, students, HR teams, agencies aur small businesses repetitive document creation ka time reduce kar sakte hain.

23. Final recommended architecture

```
Frontend: Next.js + React + TypeScript
UI: Tailwind CSS
Editor: Tiptap/Lexical
Auth: Supabase Auth
Database: Supabase PostgreSQL
Storage: Supabase Storage
AI: LLM API + structured JSON
PDF: Server-side PDF renderer
DOCX: DOCX generation library
Hosting: Vercel
Code: GitHub
Monitoring: Error + usage logging

Flow:
User → Next.js → API → AI → Structured JSON → Editor
↓
Quality/Humanize
↓
Template Renderer
↙ ↘
PDF DOCX
```

24. Final result

Is architecture ke baad aapke paas sirf ek chatbot nahi hoga. Aap ek complete AI Document Production System bana sakte hain jahan input collection, AI generation, editing, humanization, versioning, templates, document rendering aur secure export ek single workflow mein connected hon.

Build priority: Pehle MVP → real users se testing → generation quality improve → templates expand → subscriptions/API/team features add.